Vishnu Suresh

917-340-2622 | vishnups0001@gmail.com | linkedin.com/in/vishnu-suresh-08b8b5284 | github.com/vishnupsuresh

Education

The Cooper Union for the Advancement of Science and Art, New York, NY *Expected May 2029*
Bachelor of Science, Computer Science | GPA: 3.46
Coursework: Data Structures & Algorithms (C++), Computer Architecture, ECE 160 (C), CS 101 (Python), Linear Algebra
Activities: IEEE Board Member, Cooper OPS Instructor, Founder of Cooper Union Hoops

Experience

Department of Youth and Community Development, New York, NY *Jan 2026 – Present*
College Aide – IT Operations

- Built an Excel inventory tracker for 700+ laptops, desktops, phones, and monitors across 300+ employees, replacing manual logs and cutting device lookups from 15 minutes to 1 minute.
- Resolved 500+ ServiceNow tickets spanning device compliance, provisioning, and authentication across Intune/MDM, Windows Hello for Business, and SCCM/MDT.
- Imaged and deployed 100+ laptops and desktops for staff and program participants to enterprise compliance standards.

Mathnasium Learning Center, New Hyde Park, NY *Jul 2024 – Aug 2025*
Instructor

- Taught structured math problem-solving to 40+ students in grades K–12.
- Assessed skill gaps with diagnostic tests and tracked progress against individualized learning plans, helping 89% of students advance a grade level in 6 months.

Projects and Research

Custom 32-Bit Accumulator CPU | SystemVerilog, Python *Mar 2026 – May 2026*

- Designed a 32-bit accumulator-based CPU in SystemVerilog with a 3-person team, defining a 20-instruction ISA (6-bit opcode, 26-bit immediate) covering arithmetic, memory, branching, and stack-based procedure calls.
- Implemented single-cycle and pipelined datapaths sharing one control unit, with hazard detection that stalls the pipeline on stack-pointer and link-register dependencies.
- Wrote a Python assembler with label resolution and PC-relative branch encoding, translating programs (Fibonacci, factorial) into hex machine code.
- Verified both designs with testbench simulation and GTKWave waveforms, debugging timing and memory addressing.

ChargeMate: Mobile Charging Robot | Python, C++, Raspberry Pi, Linux, Arduino *Sep 2025 – Dec 2025*

- Built a client-server network of Raspberry Pis using Python sockets and multithreading, routing charge requests from button-equipped client nodes to a Raspberry Pi 5 server.
- Bridged the server to an Arduino-driven robot car via an ESP32 Wi-Fi module and UART (SoftwareSerial), translating each request into a target position for the motors.
- Assembled the robot from an ELEGOO Smart Robot Car Kit, adding 3D-printed mounts for the battery and controls.

Low-Cost EEG Brain Disorder Detection | Python, TensorFlow, Pandas *Sep 2023 – Mar 2024*

- Extracted, cleaned, and structured EEG data from public medical repositories with Pandas, resolving missing values, inconsistent sampling rates, and label formatting.
- Trained a TensorFlow CNN on the EEG data with an 80/20 train-test split, tuning batch size to reach 85% test accuracy.

Skills

Languages: Python, C++, C, Java, SystemVerilog, HTML
Data & ML: TensorFlow, PyTorch, Pandas, NumPy, Streamlit, Snowflake, MS SQL Server, Power BI, Excel
Systems & Tools: Git, Linux, GTKWave, Raspberry Pi, Arduino, ServiceNow
IT & Support: Microsoft Intune, SCCM/MDT, Windows, Microsoft 365